UNIT 9 · ARS/NET PATTERN

MCQ Bank — 70 Questions with Answers

Measurement systems, transducers, strain gauges, signal conditioning and DAQ (Q1–Q35)

Q1. The three functional stages of a measurement system are transducer, signal conditioning and:

(a) Battery
(b) Terminating/output stage
(c) Motor
(d) Pump

Q2. A device that converts a physical quantity into an electrical signal is a:

(a) Actuator
(b) Transducer
(c) Relay
(d) Rectifier

Q3. A thermocouple is an example of a/an:

(a) Passive transducer
(b) Active (self-generating) transducer
(c) Actuator
(d) Amplifier

Q4. A strain gauge is a/an:

(a) Active transducer
(b) Passive transducer (needs excitation)
(c) Self-generating
(d) Digital display

Q5. Accuracy of an instrument refers to:

(a) Repeatability
(b) Closeness to the true value
(c) Smallest division
(d) Speed

Q6. Precision refers to:

(a) Closeness to true value
(b) Repeatability of readings
(c) Range
(d) Sensitivity

Q7. The smallest change in input that an instrument can detect is its:

(a) Range
(b) Resolution
(c) Hysteresis
(d) Drift

Q8. Sensitivity is defined as:

(a) Input ÷ output
(b) Output change ÷ input change
(c) Range × span
(d) Error ÷ reading

Q9. The difference in readings for increasing vs decreasing input is:

(a) Drift
(b) Hysteresis
(c) Threshold
(d) Bias

Q10. Systematic (bias) errors are:

(a) Random and unpredictable
(b) Consistent and correctable by calibration
(c) Reduced only by averaging
(d) Never present

Q11. Random errors are best reduced by:

(a) Calibration
(b) Repeated readings and averaging
(c) Ignoring
(d) A bigger scale

Q12. The error where the instrument disturbs the measured quantity is the:

(a) Parallax error
(b) Loading effect
(c) Zero error
(d) Quantisation error

Q13. A first-order instrument's speed of response is characterised by its:

(a) Gauge factor
(b) Time constant τ
(c) Range
(d) Span

Q14. Gauge factor of a strain gauge is:

(a) $(\Delta L/L)/(\Delta R/R)$
(b) $(\Delta R/R)/(\Delta L/L)$
(c) $\Delta R \times \Delta L$
(d) R/L

Q15. Typical gauge factor of a metallic foil strain gauge is about:

(a) 0.5
(b) 2.0
(c) 20
(d) 200

Q16. Semiconductor strain gauges have gauge factors around:

(a) 2
(b) 5
(c) 20–40
(d) 500

Q17. Strain is commonly expressed in:

(a) Newtons
(b) Microstrain (10^{-6})
(c) Pascals
(d) Ohms

Q18. For GF = 2 and ε = 500 $\mu\varepsilon$, $\Delta R/R$ equals:

(a) 0.0001
(b) 0.001
(c) 0.01
(d) 0.1

Q19. The small resistance change of a strain gauge is measured using a:

(a) Transformer
(b) Wheatstone bridge
(c) Rectifier
(d) Capacitor bank

Q20. A Wheatstone bridge is balanced when:

(a) R1 = R2
(b) R1/R2 = R4/R3
(c) All R zero
(d) V_in = 0

Q21. Maximum bridge sensitivity and temperature compensation come from a:

(a) Quarter bridge
(b) Half bridge
(c) Full bridge (four active gauges)
(d) Single resistor

Q22. A dummy strain gauge is used for:

(a) Extra output
(b) Temperature compensation
(c) Calibration weight
(d) Grounding

Q23. Strain gauges are the sensing element of a:

(a) Thermometer
(b) Load cell / torque sensor
(c) Tachometer
(d) Flow meter only

Q24. Signal conditioning does NOT include:

(a) Amplification
(b) Filtering
(c) Casting
(d) Linearization

Q25. A 4–20 mA current loop is used for signal transmission because it is:

(a) Cheaper only
(b) Noise-immune over long distances
(c) Digital
(d) Wireless

Q26. Signals from a rotating shaft are carried out by:

(a) Slip rings or telemetry
(b) Gears
(c) Belts
(d) Chains

Q27. An ADC converts:

(a) Analog signal to digital
(b) Digital to analog
(c) AC to DC
(d) Force to voltage

Q28. The Nyquist criterion requires the sampling rate to be:

(a) Equal to signal frequency
(b) Greater than twice the highest signal frequency
(c) Half the frequency
(d) Any value

Q29. Sampling too slowly causes:

(a) Amplification
(b) Aliasing (false low-frequency signal)
(c) Filtering
(d) Higher accuracy

Q30. An anti-aliasing filter is placed:

(a) After the ADC
(b) Before sampling/ADC
(c) In the display
(d) Nowhere

Q31. A 12-bit ADC provides how many quantisation levels?

(a) 12
(b) 144
(c) 1024
(d) 4096

Q32. For dynamic (time-varying) measurement, the sensor's natural frequency should be:

(a) Below the signal frequency
(b) Well above the highest signal frequency
(c) Equal to it
(d) Zero

Q33. A damping ratio giving a flat, fast response without overshoot is about:

(a) 0
(b) 0.1
(c) 0.6–0.7
(d) 2.0

Q34. A piezoelectric accelerometer is typically used to measure:

(a) Temperature
(b) Vibration/acceleration
(c) Flow
(d) pH

Q35. Calibration of an instrument is done against a:

(a) Guess
(b) Known standard traceable to national standards
(c) Similar faulty unit
(d) Colour chart

Answers

Q1 — (b) Terminating/output stage
Q2 — (b) Transducer
Q3 — (b) Active (self-generating) transducer — *Active = self-generating (thermocouple, piezo).*
Q4 — (b) Passive transducer (needs excitation) — *Passive = needs excitation (strain gauge, LVDT, RTD).*
Q5 — (b) Closeness to the true value
Q6 — (b) Repeatability of readings
Q7 — (b) Resolution
Q8 — (b) Output change ÷ input change
Q9 — (b) Hysteresis
Q10 — (b) Consistent and correctable by calibration
Q11 — (b) Repeated readings and averaging
Q12 — (b) Loading effect
Q13 — (b) Time constant τ
Q14 — (b) (ΔR/R)/(ΔL/L) — *GF = (ΔR/R)/ε.*

Q15 — (b) 2.0 — *Metal ≈ 2.*
Q16 — (c) 20–40 — *Semiconductor 20–40.*
Q17 — (b) Microstrain (10^{-6})
Q18 — (b) 0.001 — *$\Delta R/R = GF\varepsilon = 2 \times 500\times10^{-6} = 0.001$.*
Q19 — (b) Wheatstone bridge
Q20 — (b) R1/R2 = R4/R3 — *Bridge balanced at R1/R2 = R4/R3.*
Q21 — (c) Full bridge (four active gauges) — *Full bridge = maximum sensitivity + temperature compensation.*
Q22 — (b) Temperature compensation — *A dummy gauge compensates temperature.*
Q23 — (b) Load cell / torque sensor
Q24 — (c) Casting
Q25 — (b) Noise-immune over long distances
Q26 — (a) Slip rings or telemetry
Q27 — (a) Analog signal to digital
Q28 — (b) Greater than twice the highest signal frequency — *Nyquist: $f_s > 2f_{max}$.*
Q29 — (b) Aliasing (false low-frequency signal) — *Too slow → aliasing.*
Q30 — (b) Before sampling/ADC — *Anti-alias filter before the ADC.*
Q31 — (d) 4096 — *12-bit = 2^{12} = 4096 levels.*
Q32 — (b) Well above the highest signal frequency — *Dynamic sensing needs natural frequency ≫ signal frequency.*
Q33 — (c) 0.6–0.7 — *Damping ~0.6–0.7 for a flat response.*
Q34 — (b) Vibration/acceleration
Q35 — (b) Known standard traceable to national standards

Temperature, pressure, force, torque, power, sound, vibration, fuel properties and flow (Q36–Q70)

Q36. A thermocouple works on the:

(a) Seebeck effect
(b) Photoelectric effect
(c) Hall effect
(d) Doppler effect

Q37. A thermocouple is a/an ___ transducer:

(a) Passive
(b) Active (self-generating)
(c) Digital
(d) Optical

Q38. The most commonly used general-purpose thermocouple is type:

(a) J
(b) K (Chromel-Alumel)
(c) T
(d) S

Q39. An RTD is usually made of:

(a) Copper only
(b) Platinum
(c) Silicon
(d) Carbon

Q40. In an RTD, as temperature rises the resistance:

(a) Falls
(b) Rises (nearly linearly)
(c) Stays constant
(d) Oscillates

Q41. A thermistor is made of a ___ and its resistance ___ with temperature (NTC):

(a) metal, rises
(b) semiconductor, falls steeply
(c) ceramic, constant
(d) gas, rises

Q42. Non-contact temperature measurement of a hot/moving surface uses a:

(a) Thermocouple
(b) Pyrometer (radiation)
(c) Manometer
(d) Rotameter

Q43. Pt100 means a platinum RTD with resistance:

(a) 100 Ω at 0 °C
(b) 100 Ω at 100 °C
(c) 1000 Ω
(d) 10 Ω

Q44. A U-tube manometer is best for measuring:

(a) Very high pressure
(b) Low pressures / differentials
(c) Temperature
(d) Flow rate only

Q45. The common mechanical pressure gauge using a curved tube that uncoils is the:

(a) Manometer
(b) Bourdon gauge
(c) Thermistor
(d) Rotameter

Q46. Dynamic (rapidly varying) pressure is best measured with a:

(a) Bourdon gauge
(b) Piezoelectric/strain transducer
(c) Manometer
(d) Dead-weight tester

Q47. Force is most commonly measured with a:

(a) Thermocouple
(b) Strain-gauge load cell
(c) Rotameter
(d) Pyrometer

Q48. A proving ring measures:

(a) Temperature
(b) Force (elastic deflection)
(c) Flow
(d) Sound

Q49. Torque on a rotating shaft is measured with:

(a) A manometer
(b) A strain-gauged torsion shaft (with slip rings/telemetry)
(c) A thermistor
(d) An orifice

Q50. Brake power from torque T and speed N (rpm) is:

(a) $2\pi NT/60000$
(b) $NT/60$
(c) T/N
(d) $9550NT$

Q51. Sound Pressure Level is defined as:

(a) $10 \log(p/p_0)$
(b) $20 \log(p/p_0)$
(c) $p - p_0$
(d) p/p_0

Q52. The reference pressure for SPL is:

(a) 1 Pa
(b) 20 μPa
(c) 1 bar
(d) 101 kPa

Q53. The A-weighting network in a sound level meter is used to:

(a) Amplify bass
(b) Match the response of the human ear (dBA)
(c) Measure flow
(d) Filter light

Q54. Vibration is most commonly sensed by a:

(a) Thermocouple
(b) Piezoelectric accelerometer
(c) Bourdon tube
(d) Rotameter

Q55. Frequency content of a vibration signal is obtained using:

(a) A manometer
(b) An FFT/spectrum analyser
(c) A calorimeter
(d) A viscometer

Q56. Calorific value of a solid or liquid fuel is measured in a:

(a) Junkers calorimeter (b) Bomb calorimeter

(c) Orsat apparatus (d) Rotameter

Q57. Calorific value of a gaseous fuel is measured with a:

(a) Bomb calorimeter (b) Junkers (gas) calorimeter

(c) Bourdon gauge (d) Pyrometer

Q58. Gross calorific value differs from net by the:

(a) Ash content (b) Latent heat of water vapour formed

(c) Sulphur (d) Density

Q59. Cetane and octane numbers are determined in a:

(a) Bomb calorimeter (b) Variable-compression CFR test engine

(c) Rotameter (d) Viscometer

Q60. Flash point of a fuel is measured by a:

(a) Redwood viscometer (b) Pensky–Martens / Abel apparatus

(c) Bomb calorimeter (d) Orsat

Q61. Viscosity of a fuel oil is measured by a:

(a) Redwood/Saybolt viscometer (b) Manometer

(c) Pyrometer (d) Load cell

Q62. Composition of a gaseous fuel/flue gas can be found by:

(a) Orsat apparatus / gas chromatograph (b) Bourdon gauge

(c) Rotameter (d) Thermistor

Q63. Head-type flow meters give flow rate proportional to:

(a) Δp (b) $\sqrt{\Delta p}$

(c) Δp^2 (d) $1/\Delta p$

Q64. Which flow meter has the LOWEST permanent pressure loss?

(a) Orifice (b) Venturi

(c) Flow nozzle (d) None

Q65. The cheapest and simplest differential-pressure flow element is the:

(a) Venturi (b) Orifice plate

(c) Coriolis (d) Turbine

Q66. A rotameter is a ___ flow meter:

(a) Differential-pressure (b) Variable-area (float in tapered tube)

(c) Positive-displacement (d) Ultrasonic

Q67. A Pitot tube measures:

(a) Mass flow (b) Point velocity (stagnation vs static pressure)

(c) Temperature (d) Density

Q68. Which meter measures MASS flow rate directly?

(a) Orifice (b) Coriolis

(c) Rotameter (d) Pitot

Q69. Fuel or water consumption is totalised by a:

(a) Positive-displacement meter (b) Pyrometer

(c) Thermocouple (d) Accelerometer

Q70. The coefficient of discharge (Cd) of a flow meter relates:

(a) Actual to ideal flow (b) Pressure to temperature

(c) Mass to volume (d) Sound to vibration

Answers

Q36 — (a) Seebeck effect — *Thermocouple — Seebeck emf.*

Q37 — (b) Active (self-generating)

Q38 — (b) K (Chromel-Alumel) — *Type K most common.*

Q39 — (b) Platinum — *RTD — platinum.*

Q40 — (b) Rises (nearly linearly) — *RTD resistance rises nearly linearly.*

Q41 — (b) semiconductor, falls steeply — *Thermistor — semiconductor, R falls steeply.*

Q42 — (b) Pyrometer (radiation) — *Pyrometer — non-contact radiation.*

Q43 — (a) 100 Ω at 0 °C — *Pt100 = 100 Ω at 0 °C.*

Q44 — (b) Low pressures / differentials

Q45 — (b) Bourdon gauge — *Bourdon C-tube uncoils under pressure.*

Q46 — (b) Piezoelectric/strain transducer

Q47 — (b) Strain-gauge load cell

Q48 — (b) Force (elastic deflection)

Q49 — (b) A strain-gauged torsion shaft (with slip rings/telemetry)

Q50 — (a) 2πNT/60000 — *BP = 2πNT/60,000 (Unit 3, Topic 12a; Unit 4, Topic 3a).*

Q51 — (b) 20 log(p/p_0) — *SPL = 20 log(p/p_0).*

Q52 — (b) 20 μPa — *p_0 = 20 μPa.*

Q53 — (b) Match the response of the human ear (dBA) — *A-weighting mimics the ear (Unit 5, Topic 6a).*

Q54 — (b) Piezoelectric accelerometer

Q55 — (b) An FFT/spectrum analyser

Q56 — (b) Bomb calorimeter — *Bomb calorimeter — solid/liquid CV.*

Q57 — (b) Junkers (gas) calorimeter — *Junkers — gas.*

Q58 — (b) Latent heat of water vapour formed — *GCV - NCV = latent heat of the water formed.*

Q59 — (b) Variable-compression CFR test engine — *Cetane/octane on a CFR engine.*

Q60 — (b) Pensky–Martens / Abel apparatus — *Flash point — Pensky–Martens.*

Q61 — (a) Redwood/Saybolt viscometer

Q62 — (a) Orsat apparatus / gas chromatograph

Q63 — (b) √Δp — *Head meters: $Q \propto \sqrt{\Delta p}$.*

Q64 — (b) Venturi — *Venturi — lowest loss.*

Q65 — (b) Orifice plate — *Orifice — cheapest.*

Q66 — (b) Variable-area (float in tapered tube) — *Rotameter = variable area.*

Q67 — (b) Point velocity (stagnation vs static pressure) — *Pitot = velocity.*

Q68 — (b) Coriolis — *Coriolis = mass flow.*

Q69 — (a) Positive-displacement meter

Q70 — (a) Actual to ideal flow